

Cardiovascular Disease Prediction Through Data Preprocessing and Classification Techniques (April 2026)

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Abstract—Cardiovascular disease is a leading cause of mortality worldwide, and early detection remains a critical challenge in healthcare. This study aims to predict cardiovascular disease by applying exploratory data analysis (EDA), preprocessing techniques, and classification models to patient data. Clinical and lifestyle variables such as blood pressure, cholesterol, glucose, and physical activity are examined to assess their predictive power. The target variable is nearly balanced, but several categorical predictors are highly skewed, requiring careful preprocessing. Several machine learning algorithms will be implemented and evaluated using standard performance metrics. Although the results are still in progress, this work highlights the importance of data preprocessing and model selection in the development of effective predictive systems for healthcare applications.

Impact Statement—Cardiovascular disease is a major global health challenge, accounting for millions of deaths annually. Early detection and prevention are critical to reducing this burden. This study explores the use of exploratory data analysis, preprocessing techniques, and machine learning classification models to predict cardiovascular disease based on clinical and lifestyle variables. The impact of this work lies in demonstrating how careful data preparation and model selection can enhance predictive accuracy.

Index Terms—Cardiovascular disease, Classification, Data preprocessing, Exploratory data analysis, Machine learning, Predictive modeling.

I. INTRODUCTION

THIS project addresses the prediction of cardiovascular disease using patient data. The dataset includes clinical and lifestyle variables such as age, blood pressure, cholesterol, glucose levels, smoking status, alcohol intake, physical activity, and BMI. Exploratory data analysis (EDA) revealed that the target variable is nearly balanced, with similar proportions of positive and negative cases. However, several categorical predictors are highly skewed, and crosstab analysis showed that higher cholesterol, glucose, and BMI are strongly associated with cardiovascular disease, while gender and smoking/alcohol intake exhibit minimal differences. These findings highlight the need for preprocessing techniques such as normalization, categorical encoding, handling of missing data, and careful treatment of skewed categorical distributions. Given that the target variable is binary, supervised learning

in the form of classification was selected as the appropriate paradigm. This choice is justified by the availability of labeled data and the clinical relevance of metrics such as recall and F1-score. Several models will be evaluated, including logistic regression, decision trees, ensemble methods, and neural networks, to determine the most effective approach for predictive healthcare applications.

II. PROBLEM DESCRIPTION

Cardiovascular disease (CVD) remains one of the leading causes of mortality worldwide. Early detection and prevention are critical to reducing morbidity and mortality rates. Traditional diagnostic methods often rely on clinical tests that are costly and time-consuming, limiting their applicability for large-scale preventive strategies. In this context, machine learning (ML) offers an opportunity to leverage patient data to build predictive models that can support clinical decision-making and promote preventive healthcare.

A. Data Base Composition

The dataset comprises approximately 70,000 patient records with 12 variables: 11 predictors and 1 binary target. Predictors include objective features (age in years, height in cm, weight in kg, gender), clinical examination results (systolic and diastolic blood pressure in mmHg, cholesterol, glucose), and self-reported lifestyle attributes (smoking, alcohol intake, physical activity). The target variable indicates the presence or absence of cardiovascular disease. No missing values were detected, therefore no imputation strategy was required. Preprocessing steps included renaming variables for clarity, converting age from days to years, and engineering an ordinal body mass index (BMI) variable with six categories (0–5) to capture weight–height relationships. The target variable was found to be nearly balanced, which facilitates predictive modeling.

Data cleaning removed unrealistic samples, including patients with height below 140 cm or above 230 cm, weight below 40 kg, systolic blood pressure outside the range of 50–250 mmHg, and diastolic blood pressure outside the range of 30–200 mmHg. Less than 3% of the dataset was discarded during this process. Most categorical variables showed weak associations with the target, yet they were retained due to the dimensionality of the problem and their potential contribution to model performance. Among numerical variables, systolic

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and diastolic blood pressure exhibited strong correlation; however, given their distinct clinical meaning and thresholds, both were preserved. Variables considered irrelevant, such as patient ID and gender, were excluded. Overall, the dataset after preprocessing provides a reliable and clinically meaningful foundation for predictive modeling of cardiovascular disease.

B. Selected ML Paradigm

Given the binary nature of the target variable and the availability of labeled data, supervised learning in the form of classification was selected as the appropriate paradigm. This choice is justified by the clinical relevance of distinguishing between patients with and without cardiovascular disease, and by the suitability of classification algorithms to model discrete outcomes. Although the target distribution is balanced, skewed predictors require careful preprocessing and feature engineering.

III. STATE OF THE ART

A. Individual Studies

Study 1 — Mridha et al. (ICDAI 2023) This work used the same Kaggle Cardiovascular Disease dataset and applied a supervised classification paradigm. Techniques included Logistic Regression, Random Forest, Gradient Boosting, and SVM, complemented with SHAP and LIME for explainability. Validation was performed using k-fold cross-validation. Metrics included Accuracy, Precision, Recall, F1-score, and ROC-AUC. Results showed that ensemble methods achieved the highest predictive performance ($AUC > 0.85$), while explainability analysis highlighted cholesterol, glucose, and BMI as key predictors [1].

Study 2 — Al-Makhadmeh & Tolba (IEEE Access, 2020) Al-Makhadmeh and Tolba [2] proposed a supervised learning paradigm for cardiovascular disease prediction, focusing on deep learning architectures. The authors implemented a hybrid CNN–LSTM model to capture both spatial and temporal patterns in patient data. Validation was performed using k-fold cross-validation to ensure robustness. The evaluation metrics included Accuracy, Sensitivity, Specificity, and ROC-AUC, with the hybrid CNN–LSTM consistently outperforming traditional classifiers such as logistic regression and decision trees. Reported results showed accuracy above 90% and AUC values exceeding 0.90, highlighting the potential of deep learning models in medical prediction tasks.

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TABLE I
COMPARISON OF METHODS, METRICS, AND RESULTS IN SELECTED STUDIES

Study	Methods	Metrics	Results
Mridha et al. (ICDAI 2023)	LR, RF, GB, SVM + XAI	Accuracy, Precision, Recall, F1, AUC	Ensembles best, $AUC > 0.85$
Al-Makhadmeh & Tolba (IEEE Access 2020)	Hybrid CNN–LSTM	Accuracy, Sensitivity, Specificity, AUC	Accuracy $> 90\%$, $AUC > 0.90$